PHYSICS 317: a challenge problem on Lagrange's equations

Challenge problem (optional): a free particle in an expanding universe. In Newtonian cosmology, space is filled with matter of uniform density $\rho(t)$, and every distance grows by one common factor: a particle that just rides the expansion sits at $r = a(t)\,x$ with x fixed, where $a(t)$ is the *scale factor* and x is its *comoving coordinate*. By Newton's shell theorem the matter inside radius r pulls a particle there as if all of it sat at the origin (and the matter outside exerts no net force), so a particle of mass m at distance r from the origin has potential energy

$$U(r,t) = \frac{2\pi G}{3}\, m\rho(t)\, r^2 \qquad \Big[\text{check: } -\frac{\partial U}{\partial r} = -\frac{Gm}{r^2}\cdot\frac{4\pi}{3}\rho r^3 = -\frac{GmM(<r)}{r^2}\Big],$$

and the scale factor obeys the Friedmann equation $\ddot{a} = -\frac{4\pi G}{3}\rho\, a$ (Newton's second law for a particle riding the expansion, with $r = ax$; you may take it as given). Now consider a particle of mass m that moves along a line through the origin but does *not* necessarily ride the expansion.

(a) Write down the Lagrangian $\mathcal{L}(r, \dot{r}, t)$. Find the Lagrange equation for r and check that it is Newton's second law with the force above. Explain why neither of our two conservation tests (an ignorable coordinate, Section 7.6; no explicit t in $\mathcal{L}$, Section 7.8) hands you a conserved quantity here.

(b) Change to the comoving coordinate $x = r/a(t)$ (a generalized coordinate whose relation to r involves t, as in Example 7.6). Show that

$$\mathcal{L} = \tfrac{1}{2}ma^2\dot{x}^2 + \frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\,x^2\Big).$$

Hint: expand the square in $\frac{1}{2}m\dot{r}^2$; write the cross term $ma\dot{a}\,x\dot{x}$ as a total time derivative minus what is left over; use the Friedmann equation on what is left over.

(c) Explain why the total time derivative can be dropped without changing the equation of motion (Problem 7.48, or argue directly from Hamilton's principle). From here on take $\mathcal{L} = \frac{1}{2}ma(t)^2\dot{x}^2$.

(d) Which coordinate is now ignorable? Find the conserved generalized momentum, and show that the *peculiar velocity* $v_{\text{pec}} \equiv a\dot{x}$, the particle's velocity relative to the expansion, falls as $1/a$. (When the universe doubles in size, a galaxy's random motion halves.)

(e) Find the Hamiltonian $\mathcal{H} = p\dot{x} - \mathcal{L}$. Is it $T+U$? Is it conserved? Use $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$ (Section 7.8) to find how $\mathcal{H}$ depends on a.

(f) General relativity gives the same conservation law for light: the momentum a local observer measures is the conserved momentum divided by a. With $p_{\text{local}} = h/\lambda$, deduce the cosmological redshift $1 + z \equiv \lambda_{\text{obs}}/\lambda_{\text{emit}} = a_{\text{obs}}/a_{\text{emit}}$.